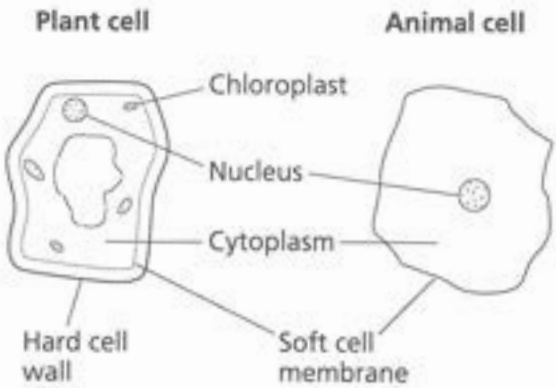
11

Structure of plant and animal cells

Many of the features of plant and animal cells are the same as the simplified diagram indicates. Plant and animal cells (excluding bacteria) contain a nucleus (housing genetic material) as well as features that synthesise protein and energy producing components. A gel-like cytoplasm is found in both cell types and the whole cell is enclosed within a cell membrane. The membrane is a thin sheet that controls what enters and leaves the cell, using a series of pores. Plants have additional features, such as chloroplasts or other plastids (photosynthetic cells), to compensate for the fact that they must produce their own foods (via photosynthesis). Lacking a skeleton, plants need another way to maintain some rigidity, so a cell wall helps to stiffen the structure, although bark also helps with this in woody species. Most bacteria have a cell wall but no cell membrane.



1 Plants are *autotrophs* and animals are *heterotrophs*. Explain what these two terms mean.

2 What is the major constituent of cell walls in plants?

3 Copy the simplified diagrams of the plant cell and animal cell. On each cell draw in one more feature not already shown and label this feature.

4 Name the part of a cell described as: '*An organelle that controls many functions of a cell and is the site of chromosome formation*'.

Eucaryotic and procaryotic cells *

Most of the cells you will encounter in biology are *eucaryotic* cells (*eu* good, *karyon* nut). Eucaryotic cells have a self-contained nucleus – the 'nut' within it – and membrane-bound organelles. Organelles are specialised parts of a cell. They include chloroplasts, Golgi bodies and mitochondria. Chloroplasts contain chlorophyll and are the site of photosynthesis. Golgi bodies are also called Golgi apparatus or just Golgi. They modify and transport proteins. Mitochondria (singular – *mitochondrion*) produce energy using cellular respiration and the chemical compound ATP (adenosine *tri*phosphate). The genetic material of eucaryotic cells is complex, with various proteins bound to the DNA and many different chromosomes.

* *Spelling variation: eukaryotic, prokaryotic.*

5 Give the full name of ATP.

6 What does this sentence describe: '*Body found in the nucleus containing the genes of the organism*'?

In contrast, *procaryotic* cells (*pro* before, *karyon* nut) are the simplest of cells. In these cells the organism's genetic information is not contained within a nucleus, nor does it have proteins bound to it. Bacteria are an example of these 'pre-nuclear' organisms. Bacteria generally contain a single chromosome. The membrane-enclosed organelles found in eucaryotes are absent in procaryotes. The nuclear material is not contained within a nucleus but is spread out randomly within the cytoplasm. Procaryotes are found in the kingdom *Monera*, which consists of primitive organisms.

7 Name two organisms found in the kingdom *Monera*.

Organelles

There is a clear division of labour among a cell's organelles. Each organelle is specialised to perform specific tasks. Organelles are suspended in a clear, colloidal, gel-like substance called *cytoplasm*. Cytoplasm is found between the outer nuclear membrane and the inner cellular membrane. (Hint: think of the children's balloons in which a smaller balloon is floating within a larger one. The cytoplasm fills the 'air space' in which the smaller balloon floats). Protoplasm consists of cytoplasm, mitochondria, nucleus and (in plants) plastids. Most organelles have a membrane (from the Latin, *membran*, meaning 'skin'), which is a thin sheet with pores (holes) covering it.

8 Give a definition for *organelle*.

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The main components of a typical plant and animal cell are listed below (✓ = present in cell).

Organelle	Animal	Plant	Function
cytoplasm (*cyto* cell)	✓	✓	gel where organelles are suspended
nucleus ('kernel')	✓	✓	houses genetic material
nucleolus ('nut')	✓	✓	within nucleus, RNA synthesis
mitochondrion (*mitos* thread)	✓	✓	energy production
Golgi apparatus*	✓	✓	exports cellular products
starch grain		✓	important food reserve for plant
vacuole ('empty space')		✓	stores water, air and other matter
endoplasmic reticulum (ER)	✓	✓	protein modification, lipid synthesis
plastid (*plast* granule)		✓	create and store various substances
chloroplast (*chloros* green)		✓	major plastid vital to photosynthesis
cell wall		✓	protects and supports cell, rigidity
cell membrane+	✓	✓	controls what enters and leaves cell
lysosome (*lysis* destroy)	✓	✓	intracellular digestion

*sometimes called *Golgi body* or just *golgi* +also called *plasma* membrane

9 What is the function of the cell wall in plants?

10 Suggest why animal cells do not have cell walls.

Research

11 The plasma membrane is said to be *selectively permeable*. Define selectively permeable and give an example.

Roles of cell components

You will be familiar with the levels of organisation in a living multicellular organism. For example, cells give rise to tissue, tissue to organs, organs to organ systems and finally, organ systems to whole organisms. Each level of organisation builds on the previous one. The cell itself is also composed of specific components, called organelles (tiny 'organs'), each of which perform clearly defined tasks. Cellular components can be identified through a light microscope by fixating (killing) the cell and staining it with dyes that will make the contents easier to see. Note: some components can be identified with a light microscope, others are so small that an electron microscope is required.

12 Name three organelles that are present in plant cells but not in animal cells.

Give scientific meanings for:

13 staining **14** unicellular **15** fixating/fixation **16** multicellular **17** protoplasm

18 Suggest why some cells are examined alive and others are killed before viewing under a microscope.

19 Why is an expensive electron microscope used to study some parts of the cell, when less expensive light microscopes are available?

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